

■ 2.6.17 Advanced Topic: Referring to Files

Many operating systems, such as UNIX, provide a number of mechanisms for referring to files without necessarily specifying their names or locations in full. *Mathematica* allows you to make use of such mechanisms in your operating system or “shell”, and also adds some mechanisms of its own.

When you refer to a file in *Mathematica*, the first thing that *Mathematica* does is to scan the name you have given to see if it contains any special “metacharacters”.

On most computer systems, *Mathematica* looks for the characters `*`, `$`, `~`, `?`, `[`, `"`, `\`, ``` and `'`. If it finds any of these characters, then *Mathematica* passes the name you have given to your operating system or shell for interpretation. If you are using the UNIX operating system, then `name*` and `$VARIABLE` constructions will be expanded at this point. Assuming that the result is a unique file, *Mathematica* will then use this file.

If you are using a file for *output*, then *Mathematica* makes no further transformations on its name: if the file does not exist, *Mathematica* will typically try to create it.

If, on the other hand, you are trying to access a file for input, say with the `<<` operator, then *Mathematica* uses a further mechanism to find the file you want.

On most computer systems, files are organized into “directories” or “folders”. *Mathematica* can search for input files in a list of directories, not necessarily just your current directory. The full specification of a file is typically *directory/name*. When you ask for a file with a particular name, *Mathematica* can search for a file with that name in a sequence of directories. The sequence of directories that *Mathematica* searches is specified by strings in the list `$Path`. (This mechanism is similar to the path mechanism used in UNIX to search for programs.)

<code>.</code>	current directory
<code>..</code>	directory one level up in the hierarchy
<code>~</code>	home directory

Special codes for specifying directories.

When you use `<<filename`, the file name you specify can contain not only alphanumeric characters, but also the metacharacters mentioned above, together with dot, slash and tilde. If it contains any other characters, however, you have to enclose it in double quotes.